



# Junlin Wu

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## EDUCATION

### Columbia University in the City of New York

Data Science | Master

Relevant Courses: Computer Vision, Computer Algorithms, Statistics and Probabilities, Machine Learning Projects with Real-World Industrial Applications

Aug 2023 - Dec 2024

New York, NY

### University of California San Diego

Data Science | Bachelor

Minor: Econ

Relevant Courses: Business Analytics and Econometrics, Machine Learning and Artificial Intelligence, Microeconomics, Macroeconomics, Data Management, Data Visualization, Data in Random Matrices, Fairness, and Algorithmic Decision-Making

Sep 2019 - Mar 2023

San Diego, CA

## PROFESSIONAL EXPERIENCE

### Columbia University Medical Center

Data Analyst

Nov 2023 - Present

New York, NY

- Participated in a live neuron recording project, analyzing neuronal activity data to correlate key metrics with animal feeding behavior and intake.
- Wrote Python scripts for data processing and analysis, quantifying the relationship between neuronal activity and feeding behavior. Visualized experimental data using Matplotlib and Seaborn, aiding the research team in gaining deeper insights into neuronal activity patterns.
- Developed automated data processing scripts, enabling researchers to automate similar data analyses in future studies.

### Beijing Zhihui Tudou Technology Co, Ltd.

Web Development

Jun 2024 - Present

Remote

- Independently developed and deployed the company website using the Vue framework. Planned and implemented subsequent development for the app, WeChat mini-program, and backend systems, optimizing user experience and improving system performance.
- Responsible for the website's daily maintenance, information updates, and optimizations to ensure stable system operation and enhanced user experience.

### Nanda Shangcheng Software, LLC.

Software Development Intern

Jun 2022 - Sep 2022

Nanjing, China

- Developed an information management system for use by the local government.
- Collaborated with clients to establish system requirements and improve usability.
- Managed project documentation and meeting notes and maintained the website.

## ACADEMIC PROJECTS

### TD Bank - Developing a Generative AI Executive Advisor to Enhance

Graduate Capstone Project

Sep 2024 - Present

New York, NY

- Developed a Generative AI Executive Advisor leveraging Retrieval-Augmented Generation (RAG) and Large Language Models (LLMs) like GPT-4 to enhance executive decision-making at TD Bank.
- Aggregated and analyzed large volumes of publicly available data, including financial reports, press releases, and market analyses.
- Created an AI system capable of summarizing critical points, identifying key issues, and suggesting pertinent questions to improve the efficiency of decision-making processes for executives.

### Chessboard Recognition and Notation Conversion

Graduate Project

Oct 2023 - Dec 2023

New York, NY

- Developed a CNN model using PyTorch and OpenCV, achieving 95% accuracy in chessboard recognition. The model can identify chessboard images and convert them into PEN-format text records.
- Utilized image processing techniques to automatically recognize and parse the board state. Future development could include integrating a chess engine to enable real-time game analysis.

### DOTA 2 Pro Match Prediction

Undergraduate Capstone Project | Team Leader

Sep 2022 - Mar 2023

San Diego, CA

- Utilized machine learning techniques to predict the outcomes of professional DOTA 2 matches, achieving effective prediction accuracy based on hero selection. The dataset covered over 30,000 matches.
- By introducing feature engineering based on hero counter-relationships, the model's accuracy was improved to 58.3%.

## EXTRACURRICULAR EXPERIENCE

### UCSD CSSA

Head of the Technology Department

Mar 2021 - Mar 2023

San Diego, CA

- Created coding tutorials for 15 team members to enhance their skills. Designed a WeChat program to facilitate class grouping for UCSD students. Developed the organisation's website using Vue.js, Vite, and Flask.
- Conducted usability tests and gathered feedback to improve program functionality. Organised academic workshops and events, including the Spring Festival Gala.

## TECHNICAL SKILLS

- **Skills:** Python (Pandas, NumPy, Matplotlib, OpenCV), Machine Learning (Scikit-learn, PyTorch, TensorFlow), Frontend (JavaScript, SOL).